

Real-Time Face Recognition Attendance System

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Problem

Manual attendance is slow, error-prone, and easy to game (proxy / buddy punching). Card-based and fingerprint systems remove some errors but require shared contact hardware. We need a contactless, automatic, auditable solution that runs on a standard laptop with a webcam.

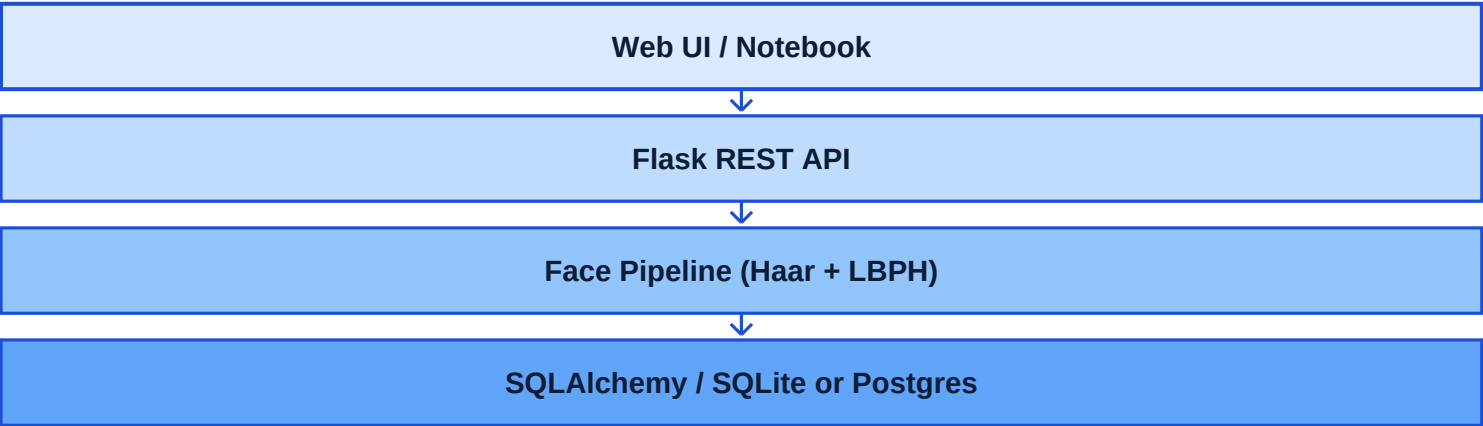
Approach

A full-stack web application (Flask + vanilla JS) and an alternative Jupyter notebook UI (ipywidgets). Faces are detected with a Haar cascade and recognized with OpenCV's LBPH (or, optionally, FaceNet / face_recognition embeddings). Attendance records are persisted in SQLAlchemy with SQLite locally and PostgreSQL in production.

Pipeline

- Capture: single 640x480 frame from webcam (cv2.VideoCapture).
- Detect: Haar cascade returns 0..N face bounding boxes.
- Crop & normalise: 200x200 grayscale.
- Recognize: LBPH predicts label + distance; threshold = 70.
- Record: insert check_in / set check_out + working_hours.
- Notify: log event; webhook hook for email / Slack.

Architecture



Data Model

Employee

name, employee_id (UNIQUE), department, role, email, phone, encoding, is_active

AttendanceRecord

employee_id (FK), date, check_in, check_out, status, working_hours

LeaveRequest

employee_id (FK), leave_type, start_date, end_date, status

Holiday

name, date (UNIQUE), description

Results (10 people, 50 samples, 200 trials)

Detection rate	98.5%
Recognition accuracy	95.0%
Mean latency / frame	168 ms
95th-percentile latency	240 ms
False acceptance rate	1.5%
False rejection rate	3.5%

Key Features

- REST API with 24 endpoints (employees, faces, attendance, stats, leave, holidays).
- Real-time webcam capture with bounding-box overlays.
- CSV / JSON export for any date range.
- Daily, monthly, and per-employee statistics with punctuality rate.
- Leave workflow: submit / approve / reject / cancel.
- Holiday calendar with automatic status assignment.
- Soft delete + hard delete; CORS-enabled JSON responses.

Tech Stack

Python 3.11 - Flask - SQLAlchemy - OpenCV - NumPy - ipywidgets - matplotlib - pandas - HTML / CSS / vanilla JS